Figure 12: Window to select the C++ class

Figure 13: New class details

```
void test::setName(QString &datastr){
    str = datastr;
}
void test::changeValue(int newValue)
{
    qDebug() << "Value " << newValue << "\nString  = " << this-
>str;
}
void test::setValue(int newValue){
    emit valueChanged(newValue);
}
```

Now, let's dissect the code. In the class test, we have added two public functions— *setValue* and *setName;* one custom signal—*valueChanged;* and two private variables —*value* and *str.* In *setValue*, the value of *int newValue* is changed, and the signal *valueChanged* is emitted, using the keyword emit. We can also pass arguments to the signal; here, we have passed the *value* variable. In *main.cpp*, we created two objects of the test class. Then we used the connect function to connect the signal *valueChanged()* emitted by *t1* to the slot *changeValue()* of *t2*. So the call to *setValue* will cause the signal emitted by *t1* to be captured by *t2*. The syntax to connect signal and slot is as follows:

```
QObject::connect(&t1,SIGNAL(valueChanged(int)),
&t2,SLOT(changeValue(int)));
```

Compile and run this sample code; you will get the following output:

```
Value  225
String  =  "manoj@nxitsolutions.com"
```

Let's cover some theory. To use signals/slots, you must inherit the class from *QObject*. The *QObject* function *connect()* creates a connection between two objects. One object will emit the signal, and the other will respond by executing its handler (slot). The *connect()* function accepts four arguments: the first object reference, the signal, the second object reference, and the slot function. A single signal can be connected to multiple slots, and a slot can be invoked for multiple signals. All the widgets and layouts too inherit from the *QObject* class. Here are some possible experiments:

1) Try with less or more arguments in the slot than the signals and see what happens.
2) Watch out for errors generated by not inheriting *QObject* class.
3) Emit the signal without connecting it to any slot

Try these to get to know more about the behaviour of signals/slot mechanisms. In the next article, we will work on the GUI classes of Qt. END

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